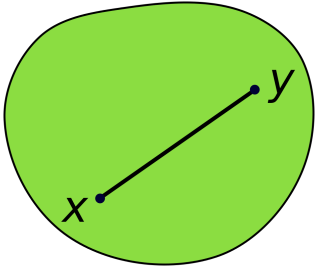
Optimization in Machine Learning

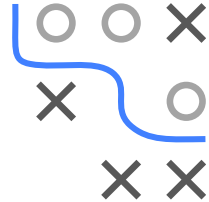
Mathematical Concepts
Convexity



Learning goals

- Convex sets
- Convex functions

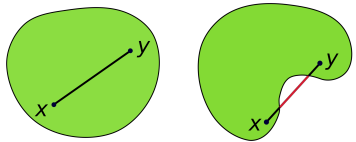
CONVEX SETS



A set of $\mathcal{S} \subseteq \mathbb{R}^d$ is **convex**, if for all $\mathbf{x}, \mathbf{y} \in \mathcal{S}$ and all $t \in [0, 1]$ the following holds:

$$\mathbf{x} + t(\mathbf{y} - \mathbf{x}) \in \mathcal{S}$$

Intuitively: Connecting line between any $\mathbf{x}, \boldsymbol{y} \in \mathcal{S}$ lies completely in $\mathcal{S}$.



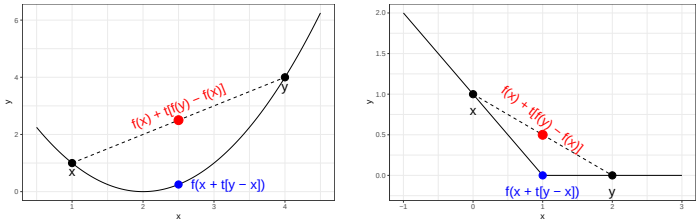
Left: convex set. **Right:** not convex. (Source: Wikipedia)

CONVEX FUNCTIONS

Let $f : \mathcal{S} \to \mathbb{R}$, $\mathcal{S}$ convex. f is **convex** if for all $\mathbf{x}, \mathbf{y} \in \mathcal{S}$ and all $t \in [0, 1]$

$$f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) \leq f(\mathbf{x}) + t(f(\mathbf{y}) - f(\mathbf{x})).$$

Intuitively: Connecting line lies above function.



Left: Strictly convex function. **Right:** Convex, but not strictly.

Strictly convex if "$<$" instead of "$\leq$". **Concave** (strictly) if the inequality holds with "$\geq$" ("$>$"), respectively.

Note: f (strictly) concave $\Leftrightarrow$ $-f$ (strictly) convex.

EXAMPLES

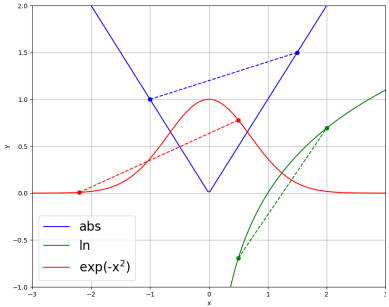
Convex function: $f(x) = |x|$

Proof:

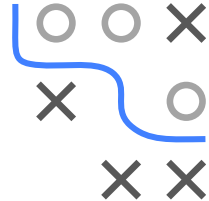
$$
\begin{aligned}
f(x + t(y - x)) &= |x + t(y - x)| = |(1 - t)x + t \cdot y| \\
&\leq |(1 - t)x| + |t \cdot y| = (1 - t)|x| + t|y| \\
&= |x| + t \cdot (|y| - |x|) = f(x) + t \cdot (f(y) - f(x))
\end{aligned}
$$

Concave function: $f(x) = \log(x)$

Neither nor: $f(x) = \exp(-x^2)$ (but log-concave)



OPERATIONS PRESERVING CONVEXITY



- **Nonnegatively weighted summation:** Weights $w_1, \ldots, w_n \geq 0$, convex functions $f_1, \ldots, f_n$: $w_1 f_1 + \cdots + w_n f_n$ also convex
  In particular: Sum of convex functions also convex
- **Composition:** g convex, f linear: $h = g \circ f$ also convex
  Proof:

$$
\begin{aligned}
h(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) &= g(f(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))) \\
&= g(f(\mathbf{x}) + t(f(\mathbf{y}) - f(\mathbf{x}))) \\
&\leq g(f(\mathbf{x})) + t(g(f(\mathbf{y})) - g(f(\mathbf{x}))) \\
&= h(\mathbf{x}) + t(h(\mathbf{y}) - h(\mathbf{x}))
\end{aligned}
$$

- **Elementwise maximization:** $f_1, \ldots, f_n$ convex functions:
  $g(\mathbf{x}) = \max\{f_1(\mathbf{x}), \ldots, f_n(\mathbf{x})\}$ also convex

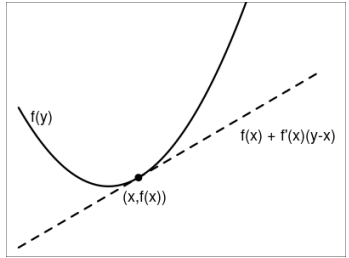
FIRST ORDER CONDITION

Prove convexity via **gradient**:
Let f be differentiable.

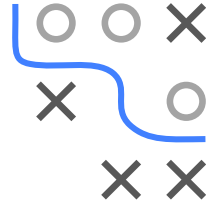
$$f \text{ (strictly) convex}$$

$$\Longleftrightarrow$$

$$f(\mathbf{y}) \overset{(>)}{\geq} f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x}) \text{ for all } \mathbf{x}, \mathbf{y} \in \mathcal{S} \text{ (s.t. } \mathbf{x} \neq \mathbf{y})$$



SECOND ORDER CONDITION



Matrix A is **positive (semi)definite** (p.(s.)d.) if $\boldsymbol{v}^T A \boldsymbol{v} \overset{(\geq)}{>} 0$ for all $\boldsymbol{v} \neq 0$.

Notation: $A \overset{(\succcurlyeq)}{\succ} 0$ for A p.(s.)d. and $B \overset{(\succcurlyeq)}{\succ} A$ if $B - A \overset{(\succcurlyeq)}{\succ} 0$

Prove convexity via **Hessian**:
Let $f \in \mathcal{C}^2$ and $H(\mathbf{x})$ be its Hessian.

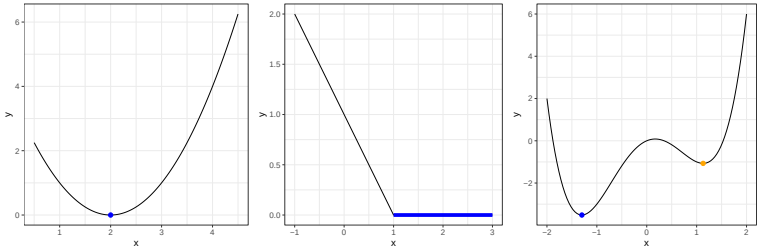
$$f \text{ (strictly) convex} \iff H(\mathbf{x}) \overset{(\succ)}{\succcurlyeq} 0 \text{ for all } \mathbf{x} \in \mathcal{S}$$

Alternatively: Since $H(\mathbf{x})$ symmetric for $f \in \mathcal{C}^2$:

$$H(\mathbf{x}) \succcurlyeq 0 \Leftrightarrow \text{all eigenvalues of } H(\mathbf{x}) \geq 0$$

CONVEX FUNCTIONS IN OPTIMIZATION

- For a convex function, every local optimum is also a global one
  $\Rightarrow$ No need for involved global optimizers, local ones are enough
- A strictly convex function has at most one optimal point
- Example for strictly convex function without optimum: $\exp$ on $\mathbb{R}$



Left: Strictly convex; exactly one local minimum, which is also global. **Middle:** Convex, but not strictly; all local optima are also global ones but not unique. **Right:** Not convex.